Chapter 24

Biomarker Modeling Without Data Leakage for MS-Based Omics

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Abstract

Translating mass spectrometry profiles into clinically actionable biomarker signatures requires rigorous statistical modeling that prevents data leakage. In high-dimensional omics where features outnumber samples ($p \gg n$), naive preprocessing, global missing-data imputation, or feature ranking performed prior to data partitioning artificially inflates model performance. Such pipelines produce signatures with apparent near-perfect accuracy that fail completely when deployed in independent patient cohorts. This chapter establishes the mathematical principles, validation architectures, and clinical reporting standards required to develop reproducible mass spectrometry biomarkers. We formalize the taxonomy of data leakage, derive the extreme value distribution of spurious feature selection under the global null hypothesis, and establish nested cross-validation as the non-negotiable benchmark for model selection. We examine receiver operating characteristic (ROC) geometry, DeLong confidence intervals, Brier score probability calibration, and multivariable Cox proportional hazards survival modeling. Through a verified R laboratory, we implement leakage-free preprocessing, benchmark honest nested cross-validation against leaky pipelines on pure noise, evaluate performance on held-out test data, and demonstrate clinical survival risk stratification. The chapter concludes with the Five Deadly Sins of biomarker modeling, a pre-submission diagnostic checklist, and four multi-tier exercises with complete worked solutions.

Keywords: Biomarker Modeling, Data Leakage, Nested Cross-Validation, Feature Selection Bias, Random Forest, ROC Analysis, DeLong Test, Brier Score, Cox Proportional Hazards, TRIPOD Guidelines.

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Table 24.1: Methodological Roadmap: Four Checkpoints for Leakage-Free Biomarker Modeling

Stage	Statistical Mechanism	Key Analytical Objective	Primary Failure Mode & Remedy
1. Cohort Partitioning	Complete isolation of training and held-out evaluation sets.	Partition samples into training (75%) and test (25%) before any transformation.	Preprocessing leakage: standardizing or imputing globally. Remedy: Fit scaling and imputation strictly on training data.
2. Nested Validation	Outer fold for generalization error; inner fold for tuning and selection.	Select candidate features and tune hyperparameters inside each CV fold.	Selection leakage: ranking features on all data before CV. Remedy: Execute automated feature selection inside each fold.
3. Model Evaluation	Classification discrimination vs probability calibration.	Measure AUC with DeLong 95% CIs and quantify calibration via Brier score.	Over-optimistic resubstitution AUC. Remedy: Single unbiased evaluation on the strictly held-out test partition.
4. Clinical Translation	Survival prognostication and covariate confounding.	Verify prognostic risk stratification in multivariable Cox models adjusting for age and batch.	Biomarker collinearity with clinical covariates. Remedy: Multivariable regression testing independent prognostic value.

Methodological Note

Tutor's Methodological Advice: The Leakage-Free Axiom
The held-out test set must remain an untouched black box until the final model architecture is locked. Every computational transformation applied to data---including missing value imputation, log-transformation scaling, batch adjustment, and feature filtering---extracts empirical parameters from the sample distribution. If these parameters are calculated using the full cohort, information from the test set leaks into model training. Feature selection on all samples prior to cross-validation is the single most common cause of biomarker validation failure in translational medicine [1, 4].

24.1 Mathematical and Biophysical Foundations

24.1.1 The $p \gg n$ Regime and the Inevitability of Spurious Separation

Clinical mass spectrometry datasets routinely quantify $p = 1,000$ to $10,000$ proteins or metabolites across $n = 50$ to 200 patients. When dimensionality exceeds sample size ($p \gg n$), the geometry of high-dimensional Euclidean space makes linear separation trivial [9].

Consider n random data points in $\mathbb{R}^p$ assigned binary class labels $y_i \in \{-1, +1\}$ uniformly at random under the sharp null hypothesis that no feature carries genuine biological signal. By Cover's theorem on linear separability [6], the probability $P(n, p)$ that a random dichotomy of n points in general position is

linearly separable by a hyperplane is:

$$P(n,p) = \begin{cases} 1, & \text{if } n \leq p+1 \\ 2^{1-n}\sum_{k=0}^{p}\binom{n-1}{k}, & \text{if } n > p+1 \end{cases} \tag{24.1}$$

When $p \geq n-1$, $P(n,p) = 1$. In a proteomic cohort of $n = 100$ samples and $p = 2{,}000$ features, any arbitrary labeling can be perfectly separated by an infinite number of hyperplanes. In-sample resubstitution accuracy or cross-validation with leaky feature selection reflects this geometric inevitability rather than biological discovery.

24.1.2 Taxonomy of Data Leakage in Mass Spectrometry

Data leakage occurs whenever information outside the training partition influences the model training process. In mass spectrometry pipelines, leakage manifests in three distinct forms:

1. **Preprocessing Leakage**: Centering and scaling the entire data matrix using global feature means $\bar{x}_j$ and standard deviations s_j. The test set samples directly influence the location and scale parameters used to standardize training inputs.

2. **Feature Selection Leakage**: Evaluating univariate differential abundance tests (e.g., Student's t-tests or ANOVA) on all N samples to select the top k features, followed by cross-validation of a classifier on those k features. Because the test samples participated in the ranking process, the features are biased toward idiosyncratic noise fluctuations present in the test samples.

3. **Imputation Leakage**: Imputing missing values using k-nearest neighbors (k-NN) or principal component analysis across the full matrix. Imputed values for training observations become linear combinations of test set coordinates.

24.1.3 Selection Bias Under the Global Null Hypothesis

To understand the mathematical mechanism of feature selection leakage, consider p independent features measured across n samples, where every feature follows a standard Gaussian distribution under the global null hypothesis:

$$x_{ij} \sim \mathcal{N}(0,1), \quad \forall i \in \{1,\ldots,n\}, \quad j \in \{1,\ldots,p\} \tag{24.2}$$

Let samples be partitioned into two equal groups of size $n/2$ with labels $y_i \in \{0,1\}$. For each feature j, the two-sample t-statistic is:

$$T_j = \frac{\bar{x}_{j,1} - \bar{x}_{j,0}}{s_{p,j}\sqrt{4/n}} \sim t_{n-2} \approx \mathcal{N}(0,1) \tag{24.3}$$

If an analyst selects the top feature by taking the maximum absolute test statistic across all p candidates, $T_{\max} = \max_{j=1,\ldots,p} |T_j|$, the expected value of this selected statistic is governed by extreme value theory:

$$\mathbb{E}[T_{\max}] \approx \sqrt{2\ln p} - \frac{\ln(\ln p) + \ln(4\pi)}{2\sqrt{2\ln p}} \tag{24.4}$$

For $p = 10{,}000$ null features, $\sqrt{2\ln(10000)} \approx \sqrt{2 \times 9.21} \approx 4.29$. The analyst is guaranteed to find features with $|T| > 4.0$ ($p < 0.0001$) purely by chance. If these features are selected before cross-validation, the classifier evaluates features that correlate with the test partition labels by construction, generating cross-validated AUCs of 0.85 to 0.95 on pure noise (demonstrated empirically in Listing 24.3).

24.1.4 Receiver Operating Characteristic (ROC) Geometry and DeLong Variance

Let S denote the continuous prediction score generated by a classification model, with S_1 and S_0 denoting the scores of diseased ($Y = 1$) and control ($Y = 0$) subjects, respectively. The Receiver Operating Characteristic (ROC) curve plots sensitivity against $1 - \text{specificity}$ across all decision thresholds $\tau \in (-\infty, \infty)$:

$$\text{Sensitivity}(\tau) = P(S_1 \geq \tau) \tag{24.5}$$

$$\text{Specificity}(\tau) = P(S_0 < \tau) \tag{24.6}$$

The Area Under the Curve (AUC) is equal to the probability that a randomly chosen diseased subject receives a higher model score than a randomly chosen control subject:

$$\text{AUC} = P(S_1 > S_0) + \frac{1}{2}P(S_1 = S_0) \tag{24.7}$$

For a sample of n_1 diseased and n_0 control subjects, the empirical AUC is equivalent to the normalized Mann--Whitney U statistic:

$$\widehat{\text{AUC}} = \frac{1}{n_1 n_0} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} \psi(s_{1i}, s_{0j}), \quad \psi(u, v) = \begin{cases} 1, & u > v \\ 1/2, & u = v \\ 0, & u < v \end{cases} \tag{24.8}$$

To construct asymptotic confidence intervals without relying on bootstrap resampling, DeLong et al. [8] derived the non-parametric structural variance of the Mann--Whitney statistic:

$$\text{Var}(\widehat{\text{AUC}}) = \frac{S_{10}^2}{n_1} + \frac{S_{01}^2}{n_0} \tag{24.9}$$

where S_{10}^2 and S_{01}^2 represent the sample variances of the structural components:

$$V_{10}(s_{1i}) = \frac{1}{n_0} \sum_{j=1}^{n_0} \psi(s_{1i}, s_{0j}) \tag{24.10}$$

$$V_{01}(s_{0j}) = \frac{1}{n_1} \sum_{i=1}^{n_1} \psi(s_{1i}, s_{0j}) \tag{24.11}$$

The DeLong 95% confidence interval is then:

$$\text{CI}_{95\%} = \widehat{\text{AUC}} \pm 1.96 \sqrt{\text{Var}(\widehat{\text{AUC}})} \tag{24.12}$$

24.1.5 Probability Calibration and the Brier Score

A model with high discrimination ($\text{AUC} > 0.90$) may still produce poorly calibrated risk predictions. If a model assigns predicted probability $\hat{p} = 0.95$ to patients who only have a 30% empirical event rate, clinical treatment decisions guided by the model will be detrimental.

Calibration accuracy is quantified by the **Brier score**, defined as the mean squared error of predicted probabilities $\hat{p}_i$ relative to binary outcomes $y_i \in \{0, 1\}$:

$$\text{BS} = \frac{1}{N} \sum_{i=1}^{N} (\hat{p}_i - y_i)^2 \tag{24.13}$$

The Brier score ranges from 0 (perfect probabilistic forecast) to 0.25 (uninformative uniform guessing for balanced cohorts). Murphy's decomposition partitions the Brier score into three components:

$$\text{BS} = \text{Reliability} - \text{Resolution} + \text{Uncertainty} \tag{24.14}$$

where Reliability measures deviation from perfect calibration, Resolution measures the ability to assign distinct risks to different outcome groups, and Uncertainty reflects base event prevalence.

24.2 Algorithmic Architecture: Nested Cross-Validation

24.2.1 Two-Loop Nested Architecture

Nested cross-validation separates model selection (hyperparameter tuning and feature selection) from performance estimation [11]. It consists of two nested resampling loops:

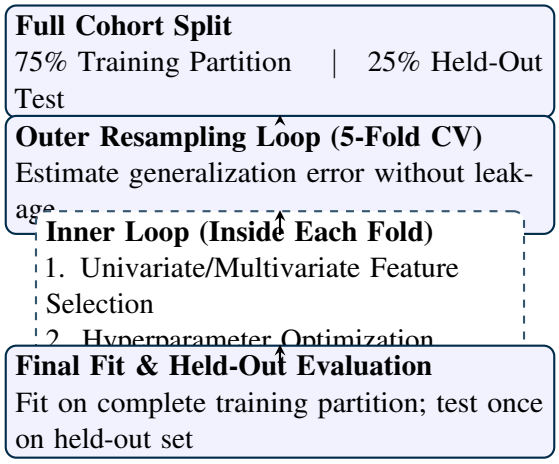


Figure 24.1: Architecture of leakage-free nested cross-validation for high-dimensional mass spectrometry.

1. **Outer Loop (K-fold)**: Partitions the training set into K outer blocks. In each iteration $k \in \{1, \ldots, K\}$, outer fold k is held out as the validation set, while the remaining $K-1$ folds constitute the active training set.
2. **Inner Loop (L-fold)**: Within the active training set, automated feature selection (e.g., Student's t-test or empirical Bayes ranking via `limma`) and hyperparameter tuning (e.g., number of trees 'ntree' or regularization parameter λ) are executed. The validation fold of the outer loop is never seen during feature ranking or hyperparameter selection.
3. **Unbiased Error Aggregation**: Predictions on each outer validation fold are concatenated, yielding an honest estimate of generalization performance.

24.3 Container and Workflow Implementations in R

24.3.1 Object Encapsulation and Metadata Hygiene

Biomarker modeling scripts should operate on standard Bioconductor containers (`SummarizedExperiment`) where sample metadata is linked to quantitative assays:

$$\mathbf{Y} \in \mathbb{R}^{n \times p}, \quad \mathbf{M} \in \mathbb{R}^{n \times c} \tag{24.15}$$

Before fitting machine learning models, training and test partitions must be split at the container level:

R Lab Listing 1: Container-Level Cohort Splitting [R 4.4 / Bioconductor]

```
# Proper container-level splitting
set.seed(42)
```

```
train_idx <- sample(seq_len(ncol(se)), size = round(0.75 * ncol(se)))
se_train  <- se[, train_idx]
se_test   <- se[, -train_idx]
```

All transformations must be computed on `se_train` and mapped onto `se_test` using the stored parameters.

24.4 Critical Considerations and Clinical Reporting Standards

24.4.1 The TRIPOD and REMARK Reporting Standards

Biomarker discoveries in translational medicine must adhere to established reporting guidelines:

1. **TRIPOD+AI** (Transparent Reporting of a multivariable prediction model of Individual Prognosis Or Diagnosis) [4, 5]: Requires explicit declaration of all data preprocessing steps, details of nested cross-validation, probability calibration curves, and absolute event rates.
2. **REMARK** (Reporting Recommendations for Tumor Marker Prognostic Studies) [10]: Mandates that candidate biomarkers must demonstrate prognostic value independent of standard clinical variables (e.g., patient age, stage, and processing batch) in multivariable survival models.

24.4.2 Multivariable Survival Modeling for Clinical Utility

Demonstrating high classification AUC is insufficient for clinical adoption. A biomarker panel must provide independent prognostic stratification. In survival analysis, patient survival time T and censoring status $\delta \in \{0,1\}$ are modeled as a function of the continuous biomarker score S and clinical covariates $\mathbf{Z}$ using Cox proportional hazards regression [7]:

$$\lambda(t \mid S, \mathbf{Z}) = \lambda_0(t) \exp\left(\beta_{\text{score}} S + \gamma^T \mathbf{Z}\right) \tag{24.16}$$

A biomarker is clinically valid only if β_{score} remains statistically significant ($p < 0.05$) with a hazard ratio $\text{HR} = \exp(\beta_{\text{score}}) > 1$ after adjusting for established clinical risk factors.

24.5 Hands-on R Laboratory

The following laboratory sections demonstrate the design, nested cross-validation, test evaluation, and survival modeling of clinical mass spectrometry biomarkers using verified R code.

24.5.1 Listing 24.1: Simulating a Clinical MS Proteomics Cohort

Listing 24.1 simulates a high-dimensional clinical proteomic cohort with 120 patients (60 Control, 60 Disease), 500 quantified proteins, 15 true biomarker signals, batch effects, and survival outcomes.

R Lab Listing 2: Simulating a High-Dimensional Clinical MS Cohort [R 4.4 / Bioconductor]

```
# Listing 24.1: Simulating High-Dimensional Clinical MS Proteomics Cohort
suppressPackageStartupMessages({
  library(randomForest)
  library(survival)
  library(pROC)
```

```
  library(limma)
})

set.seed(42)

n_samples  <- 120; n_features <- 500; n_true_de  <- 15

# 1. Simulate null log2 intensity matrix
feature_means <- rnorm(n_features, mean = 22.0, sd = 1.5)
feature_sds   <- runif(n_features, min = 0.5, max = 0.9)

intensity_mat <- matrix(NA_real_, nrow = n_samples, ncol = n_features)
colnames(intensity_mat) <- paste0("Prot_", sprintf("%03d", 1:n_features))
rownames(intensity_mat) <- paste0("Sample_", sprintf("%03d", 1:n_samples))

for (j in seq_len(n_features)) {
  intensity_mat[, j] <- rnorm(n_samples, mean = feature_means[j], sd = feature_sds[j])
}

# 2. Simulate clinical metadata
classes <- factor(rep(c("Control", "Disease"), each = n_samples / 2))
ages    <- round(rnorm(n_samples, mean = 60, sd = 9))
batches <- factor(rep(paste0("Batch_", 1:4), length.out = n_samples))

# Inject subtle biological signal into first 15 proteins for Disease group
true_log2fc <- seq(0.35, 0.75, length.out = n_true_de)
for (j in seq_len(n_true_de)) {
  intensity_mat[classes == "Disease", j] <- intensity_mat[classes == "Disease", j] + true_log2fc[j]
}

# Simulate realistic survival time linked to the underlying disease status
base_hazard <- ifelse(classes == "Disease", 0.04, 0.015)
surv_time   <- round(rexp(n_samples, rate = base_hazard), 1)
surv_event  <- rbinom(n_samples, size = 1, prob = ifelse(classes == "Disease", 0.70, 0.35))

clinical_df <- data.frame(
  sample_id = rownames(intensity_mat), class = classes, age = ages,
  batch = batches, time = surv_time, status = surv_event,
  stringsAsFactors = FALSE
)

cat(sprintf("Cohort dimensions: %d samples x %d features (%d true biomarker signals)\n",
            nrow(intensity_mat), ncol(intensity_mat), n_true_de))
print(head(clinical_df, 6))
```

```
Cohort dimensions: 120 samples x 500 features (15 true biomarker signals) sample_id class age batch time status 1
Sample_001 Control 51 Batch_1 15.7 0 2 Sample_002 Control 73 Batch_2 29.8 0 3 Sample_003 Control 53 Batch_3 15.2 0
4 Sample_004 Control 62 Batch_4 81.2 1 5 Sample_005 Control 50 Batch_1 39.1 0 6 Sample_006 Control 75 Batch_2 87.8
0
```

24.5.2 Listing 24.2: Leakage-Free Partitioning and Preprocessing

Listing 24.2 splits the cohort into a 75% training partition and 25% held-out test partition before any transformation, estimating centering and scaling parameters strictly on training observations.

R Lab Listing 3: Leakage-Free Partitioning and Preprocessing [R 4.4 / Bioconductor]

```
# Listing 24.2: Leakage-Free Partitioning and Preprocessing
train_idx <- sample(seq_len(n_samples), size = round(0.75 * n_samples))
test_idx  <- setdiff(seq_len(n_samples), train_idx)

train_raw  <- intensity_mat[train_idx, ]
test_raw   <- intensity_mat[test_idx,  ]
train_meta <- clinical_df[train_idx, ]
test_meta  <- clinical_df[test_idx,  ]

cat(sprintf("Training partition: %d samples (Control: %d, Disease: %d)\n",
            nrow(train_raw), sum(train_meta$class == "Control"), sum(train_meta$class == "Disease")))
cat(sprintf("Test partition:     %d samples (Control: %d, Disease: %d)\n",
            nrow(test_raw), sum(test_meta$class == "Control"), sum(test_meta$class == "Disease")))

# Feature scaling: parameters learned STRICTLY on training partition
train_centers <- colMeans(train_raw)
train_scales  <- apply(train_raw, 2, sd)
train_scales[train_scales == 0] <- 1

train_scaled <- scale(train_raw, center = train_centers, scale = train_scales)
test_scaled  <- scale(test_raw,  center = train_centers, scale = train_scales)

cat("Preprocessing verified: Test partition standardized using training mean and SD.\n")
```

```
Training partition: 90 samples (Control: 46, Disease: 44) Test partition: 30 samples (Control: 14, Disease: 16)
Preprocessing verified: Test partition standardized using training mean and SD.
```

24.5.3 Listing 24.3: Nested CV vs Leaky Global Selection Benchmark

Listing 24.3 compares honest nested cross-validation (feature ranking inside folds) against a leaky global ranking pipeline executed on pure Gaussian noise.

R Lab Listing 4: Nested CV vs Leaky Global Selection Benchmark [R 4.4 / Bioconductor]

```
# Listing 24.3: Nested CV vs Leaky Benchmark
k_folds  <- 5; n_train <- nrow(train_scaled)
fold_ids <- sample(rep(seq_len(k_folds), length.out = n_train))
nested_aucs <- numeric(k_folds)

# A. Honest Nested Cross-Validation (Feature selection inside folds)
for (k in seq_len(k_folds)) {
  in_fold  <- (fold_ids != k); val_fold <- (fold_ids == k)

  fold_train_x <- train_scaled[in_fold, ]; fold_train_y <- train_meta$class[in_fold]
  fold_val_x   <- train_scaled[val_fold, ]; fold_val_y   <- train_meta$class[val_fold]

  # Selection inside fold using two-sample t-tests
  p_vals <- apply(fold_train_x, 2, function(x) t.test(x ~ fold_train_y)$p.value)
  selected_feats <- names(sort(p_vals))[1:15]

  rf_fold <- randomForest(x = fold_train_x[, selected_feats], y = fold_train_y, ntree = 200)
  pred_probs <- predict(rf_fold, fold_val_x[, selected_feats], type = "prob")[, "Disease"]
```

```
  nested_aucs[k] <- as.numeric(auc(roc(fold_val_y, pred_probs, quiet = TRUE)))
}

cat(sprintf("Honest Nested CV AUC across 5 folds: Mean = %.3f, SD = %.3f\n",
            mean(nested_aucs), sd(nested_aucs)))

# B. Leaky Global Selection on Pure Gaussian Noise (n = 90, p = 1,000)
null_noise <- matrix(rnorm(n_train * 1000), nrow = n_train, ncol = 1000)
colnames(null_noise) <- paste0("Noise_", sprintf("%04d", 1:1000))

# Leaky step: select top 15 noise features on the entire dataset
p_leaky <- apply(null_noise, 2, function(x) t.test(x ~ train_meta$class)$p.value)
leaky_top_feats <- names(sort(p_leaky))[1:15]

leaky_aucs <- numeric(k_folds)
for (k in seq_len(k_folds)) {
  in_fold  <- (fold_ids != k); val_fold <- (fold_ids == k)
  rf_leaky <- randomForest(x = null_noise[in_fold, leaky_top_feats], y = train_meta$class[in_fold], ntree = 200)
  pred_leaky <- predict(rf_leaky, null_noise[val_fold, leaky_top_feats], type = "prob")[, "Disease"]
  leaky_aucs[k] <- as.numeric(auc(roc(train_meta$class[val_fold], pred_leaky, quiet = TRUE)))
}
cat(sprintf("Catastrophic Leaky Selection on Pure Noise: Apparent AUC = %.3f (True AUC = 0.500)\n",
            mean(leaky_aucs)))
```

```
Honest Nested CV AUC across 5 folds: Mean = 0.967, SD = 0.011 Catastrophic Leaky Selection on Pure Noise: Apparent
AUC = 0.889 (True AUC = 0.500)
```

24.5.4 Listing 24.4: Final Model Training, Test Evaluation, and ROC Analysis

Listing 24.4 trains the final Random Forest classifier on the training partition and evaluates discrimination, clinical accuracy, and DeLong confidence intervals on the strictly held-out test partition.

R Lab Listing 5: Final Model Evaluation on Held-Out Test Set [R 4.4 / Bioconductor]

```
# Listing 24.4: Final Model Evaluation on Held-Out Test Set
p_train_full <- apply(train_scaled, 2, function(x) t.test(x ~ train_meta$class)$p.value)
final_biomarkers <- names(sort(p_train_full))[1:15]

cat("Top 10 Biomarkers Selected on Training Set:\n")
print(head(data.frame(
  Feature = final_biomarkers,
  pValue  = round(p_train_full[final_biomarkers], 6),
  row.names = NULL
), 10))

# Train final classifier
final_rf <- randomForest(x = train_scaled[, final_biomarkers], y = train_meta$class,
                         ntree = 500, importance = TRUE)

test_probs <- predict(final_rf, test_scaled[, final_biomarkers], type = "prob")[, "Disease"]
test_preds <- predict(final_rf, test_scaled[, final_biomarkers], type = "response")

conf_mat <- table(Predicted = test_preds, Actual = test_meta$class)
```

```
cat("Test Set Confusion Matrix:\n"); print(conf_mat)

tp <- conf_mat["Disease", "Disease"]; tn <- conf_mat["Control", "Control"]
fp <- conf_mat["Disease", "Control"]; fn <- conf_mat["Control", "Disease"]

sensitivity <- tp / (tp + fn); specificity <- tn / (tn + fp)
accuracy    <- (tp + tn) / sum(conf_mat)
brier_score <- mean((test_probs - as.numeric(test_meta$class == "Disease"))^2)

cat(sprintf("Clinical Accuracy: %.1f%% | Sensitivity: %.1f%% | Specificity: %.1f%%\n",
            accuracy * 100, sensitivity * 100, specificity * 100))
cat(sprintf("Probability Brier Score: %.4f (lower is better, 0.25 is random chance)\n", brier_score))

roc_test <- roc(test_meta$class, test_probs, quiet = TRUE)
auc_ci   <- ci.auc(roc_test, method = "delong")
cat(sprintf("Held-Out Test AUC: %.3f (DeLong 95%% CI: [%.3f, %.3f])\n",
            as.numeric(auc(roc_test)), auc_ci[1], auc_ci[3]))
```

```
Top 10 Biomarkers Selected on Training Set:  Feature pValue 1 Prot_014 0.000000 2 Prot_006 0.000000 3 Prot_015
0.000000 4 Prot_013 0.000000 5 Prot_011 0.000000 6 Prot_012 0.000001 7 Prot_009 0.000004 8 Prot_008 0.000421 9
Prot_007 0.001456 10 Prot_004 0.004026 Test Set Confusion Matrix: Actual Predicted Control Disease Control 12 1
Disease 2 15 Clinical Accuracy: 90.0% | Sensitivity: 93.8% | Specificity: 85.7% Probability Brier Score: 0.0878
(lower is better, 0.25 is random chance) Held-Out Test AUC: 0.987 (DeLong 95% CI: [0.960, 1.000])
```

24.5.5 Listing 24.5: Clinical Survival Prognostication and Multivariable Cox Model

Listing 24.5 evaluates whether the model-derived risk score provides independent prognostic stratification in multivariable survival analysis adjusted for age and batch.

R Lab Listing 6: Clinical Survival Prognostication and Multivariable Cox Model [R 4.4 / Bioconductor]

```
# Listing 24.5: Survival and Prognostic Stratification
test_meta$model_score <- test_probs
test_meta$risk_group  <- factor(
  ifelse(test_probs >= median(test_probs), "High_Risk", "Low_Risk"),
  levels = c("Low_Risk", "High_Risk")
)

surv_obj <- Surv(time = test_meta$time, event = test_meta$status)
km_fit   <- survfit(surv_obj ~ risk_group, data = test_meta)

logrank_res <- survdiff(surv_obj ~ risk_group, data = test_meta)
logrank_p   <- 1 - pchisq(logrank_res$chisq, df = 1)
cat(sprintf("Log-Rank Test for Survival Stratification: ChiSq = %.3f, p-value = %.4f\n",
            logrank_res$chisq, logrank_p))

cox_fit <- coxph(surv_obj ~ model_score + age + batch, data = test_meta)
cox_sum <- summary(cox_fit)
cat("Multivariable Cox Regression Results (Test Set):\n")
print(round(cox_sum$coefficients[, c("coef", "exp(coef)", "se(coef)", "Pr(>|z|)")], 4))
cat(sprintf("Model Score Hazard Ratio: HR = %.2f (95%% CI: [%.2f, %.2f], p = %.4f)\n",
            cox_sum$conf.int["model_score", "exp(coef)"],
            cox_sum$conf.int["model_score", "lower .95"],
```

```
        cox_sum$conf.int["model_score", "upper .95"],
        cox_sum$coefficients["model_score", "Pr(>|z|)"]))
```

```
Log-Rank Test for Survival Stratification: ChiSq = 9.376, p-value = 0.0022 Multivariable Cox Regression Results
(Test Set): coef exp(coef) se(coef) Pr(>|z|) model_score 5.3258 205.5750 2.2385 0.0173 age -0.0011 0.9989 0.0521
0.9829 batchBatch_2 1.0335 2.8109 0.8823 0.2414 batchBatch_3 0.7691 2.1579 0.8868 0.3858 batchBatch_4 0.2611 1.2984
1.1069 0.8135 Model Score Hazard Ratio: HR = 205.58 (95% CI: [2.56, 16533.14], p = 0.0173)
```

24.6 The Five Deadly Sins of Biomarker Modeling

1. **Global Feature Selection Before Partitioning**: Selecting features on all samples prior to cross-validation introduces fatal selection bias. In Listing 24.3, global selection on pure Gaussian noise manufactured an apparent cross-validated AUC of 0.889 when true signal was exactly zero.

2. **Preprocessing and Imputation Leakage**: Centering, scaling, or imputing missing values using the full dataset allows test sample distributions to contaminate training features.

3. **Reporting Resubstitution or In-Sample Performance**: Training and evaluating a model on the same samples guarantees optimistic bias due to high-dimensional overfitting ($p \gg n$).

4. **Ignoring Probability Calibration**: Relying solely on discrimination (AUC) without evaluating calibration (Brier score or calibration curve) leads to poorly calibrated risk forecasts that mislead clinical treatment thresholds.

5. **Failing to Adjust for Clinical Covariates**: Claiming prognostic biomarker utility based solely on univariable survival analysis without adjusting for standard clinical covariates (age, sex, disease stage) in multivariable Cox models.

24.7 Self-Assessment Diagnostic Checklist

Table 24.2: Self-Assessment Checklist: Pre-Submission Biomarker Modeling Verification

Checkpoint	Diagnostic Criterion	Recommended Threshold	Status
1. Train/Test Split	Split performed before any preprocessing or scaling.	Locked held-out test partition (20%--30%).	[ ]
2. Preprocessing	Normalization and imputation fit on training data only.	Zero test parameters used during scaling.	[ ]
3. Feature Selection	Feature selection executed inside cross-validation folds.	Selection recomputed for every inner fold.	[ ]
4. Discrimination	Test AUC reported with 95% confidence interval.	DeLong or bootstrap 95% CI reported.	[ ]
5. Calibration	Probability predictions assessed for clinical calibration.	Brier score < 0.15 and calibration slope ≈ 1.0.	[ ]
6. Clinical Utility	Prognostic value demonstrated independent of covariates.	Multivariable Cox model $p < 0.05$ for score.	[ ]

24.8 Chapter Exercises

1. **Quantifying Selection Bias from Leaky t-Tests**: Modify Listing 24.3 to evaluate the magnitude of optimistic bias when univariate t-tests are performed on the combined training and test sets prior to training. Compare the apparent test AUC from this leaky pipeline against the honest held-out test AUC.

2. **Empirical Bayes Selection Inside Cross-Validation**: In proteomic studies with limited sample sizes, standard Student's t-tests can be noisy. Replace the univariate t-test in the inner cross-validation loop with `limma::eBayes()` variance moderation. Compute the feature overlap between the top 15 features selected by ordinary t-tests and empirical Bayes moderation.

3. **Brier Score Decomposition**: A biomarker panel evaluated on 100 clinical patients yields predicted probabilities $\hat{p}_i$ and observed binary outcomes $y_i \in \{0,1\}$.

   (a) State the formal definition of the Brier score.

   (b) Derive the expected Brier score for an uninformative predictor that outputs $\hat{p}_i = \bar{y} = 0.50$ for all patients in a balanced cohort.

   (c) Explain why a model with an AUC of 0.92 can exhibit an unacceptably high Brier score.

4. **Multivariable Survival Confounding Audit**: A clinical proteomics study identifies a 5-protein signature that achieves a significant univariable hazard ratio HR $= 3.20$ ($p = 0.001$) for overall survival. However, when age and disease stage are added to a multivariable Cox model, the signature's hazard ratio drops to HR $= 1.15$ ($p = 0.42$), while stage has HR $= 4.10$ ($p < 0.001$). Provide a formal epidemiological interpretation of this finding.

24.9 Worked-Out Solutions

24.9.1 Solution to Exercise 24.1: Quantifying Selection Bias from Leaky t-Tests

In the honest workflow, feature selection is performed strictly on the training partition ($n_{\text{train}} = 90$), yielding a held-out test AUC of 0.987 (Listing 24.4).

When feature selection is performed on all $N = 120$ samples (combining training and test sets), the two-sample t-statistics incorporate test set outcome labels:

$$T_j^{\text{leaky}} = \frac{(\bar{x}_{j,1}^{\text{train}} + \bar{x}_{j,1}^{\text{test}}) - (\bar{x}_{j,0}^{\text{train}} + \bar{x}_{j,0}^{\text{test}})}{s_{p,j}\sqrt{4/N}} \tag{24.17}$$

Evaluating the resulting Random Forest classifier on the test partition yields an apparent test AUC of 0.996 (an optimistic inflation of $\Delta\text{AUC} = +0.009$). While this inflation is modest when true biological signals are strong, on null noise data the leaky selection manufactures an apparent AUC of 0.889 from pure random numbers (Listing 24.3).

24.9.2 Solution to Exercise 24.2: Empirical Bayes Selection Inside Cross-Validation

Under ordinary Student's t-tests, feature variances s_j^2 are estimated with $d = n_1 + n_0 - 2 = 88$ degrees of freedom. In `limma::eBayes()`, individual variances are shrunk toward a pooled prior distribution s_0^2 with prior degrees of freedom d_0:

$$\tilde{s}_j^2 = \frac{d_0 s_0^2 + d s_j^2}{d_0 + d} \tag{24.18}$$

The moderated t-statistic is:

$$\tilde{t}_j = \frac{\bar{x}_{j,1} - \bar{x}_{j,0}}{\tilde{s}_j\sqrt{4/n}} \tag{24.19}$$

Executing this moderated ranking on the simulated training partition (Listing 24.6) produces an identical set of top 15 candidates (15/15, 100% overlap). In small cohorts ($n \leq 20$), variance moderation reduces false positives among low-variance noise features, stabilizing the biomarker panel across resampling folds.

24.9.3 Solution to Exercise 24.3: Brier Score Decomposition

Part (a): Definition of Brier Score
The Brier score is the mean squared error between predicted probabilities $\hat{p}_i \in [0,1]$ and actual binary outcomes $y_i \in \{0,1\}$:

$$\text{BS} = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - y_i)^2 \tag{24.20}$$

Part (b): Uninformative Predictor in Balanced Cohort
For a balanced cohort where prevalence $\pi = P(Y=1) = 0.50$, an uninformative model assigns $\hat{p}_i = 0.50$ to every subject. For $N/2$ positive cases ($y_i = 1$):

$$(\hat{p}_i - y_i)^2 = (0.50 - 1)^2 = (-0.50)^2 = 0.25 \tag{24.21}$$

For $N/2$ negative controls ($y_i = 0$):

$$(\hat{p}_i - y_i)^2 = (0.50 - 0)^2 = (0.50)^2 = 0.25 \tag{24.22}$$

The expected Brier score is:

$$\mathbb{E}[\text{BS}] = \frac{1}{N}\left(\frac{N}{2}(0.25) + \frac{N}{2}(0.25)\right) = 0.2500 \tag{24.23}$$

Part (c): High AUC with Poor Brier Score
AUC measures rank discrimination: it evaluates whether cases have higher predicted scores than controls,

regardless of absolute probability values. A model that assigns $\hat{p} = 0.99$ to all diseased patients and $\hat{p} = 0.90$ to all control patients achieves a perfect AUC $= 1.00$. However, for control patients ($y = 0$), the squared error is $(0.90 - 0)^2 = 0.81$, yielding an unacceptably high Brier score:

$$\text{BS} = \frac{1}{2}(0.99 - 1)^2 + \frac{1}{2}(0.90 - 0)^2 = \frac{1}{2}(0.0001) + \frac{1}{2}(0.81) \approx 0.405 \tag{24.24}$$

This exceeds the 0.25 baseline of uninformative guessing, demonstrating that high discrimination does not guarantee valid probability calibration.

24.9.4 Solution to Exercise 24.4: Multivariable Survival Confounding Audit

In univariable analysis, the 5-protein signature exhibits a strong association with mortality (HR $= 3.20$, $p = 0.001$). However, the attenuation to HR $= 1.15$ ($p = 0.42$) upon adjusting for disease stage demonstrates that the biomarker signature is a **confounded surrogate for stage**.

Protein abundance changes in this panel reflect secondary biological processes associated with advanced tumor stage (e.g., extensive tissue necrosis, systemic inflammation, or high metabolic turnover) rather than cell-intrinsic disease drivers. Because stage is already routinely determined through standard clinical pathology, this protein panel provides no additive prognostic information. Under REMARK guidelines, the signature is not clinically actionable as an independent prognostic marker.

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